



Session 14C: Higher Derivatives — Patterns, Concavity, and the n -th Derivative

Phase 2 — Classical Techniques | 60 min

Differentiate once — you get slope. Differentiate again — you get curvature. Differentiate n times — you find a pattern. This session turns repeated differentiation from tedious computation into pattern recognition.

Prerequisites: 14A (basic derivative dictionary), 14B (product/quotient/chain/implicit)

Part A: Higher Derivatives — Just Keep Differentiating

The procedure: Take the derivative. Then take the derivative of that. Repeat until you reach the requested order. There is no new rule — only persistence.

Example 1: Computing f'' and f''' — No New Rules

Notation: $f'(x) = \frac{dy}{dx}$ (first). $f''(x) = \frac{d^2y}{dx^2}$ (second). $f'''(x) = \frac{d^3y}{dx^3}$ (third). $f^{(n)}(x) = \frac{d^ny}{dx^n}$ (n -th, for $n \geq 4$).

Procedure:

Step	Action
1	Compute $f'(x)$ using the dictionary + rules from 14A/14B

Step	Action
2	Compute $f''(x)$ by differentiating $f'(x)$ — apply all rules again
3	Compute $f'''(x)$ by differentiating $f''(x)$
...	Repeat until the requested order

$$f(x) = x^4 - 3x^3 + 2x:$$

Order	Derivative	How
$f'(x)$	$4x^3 - 9x^2 + 2$	Power rule on each term
$f''(x)$	$12x^2 - 18x$	Differentiate $4x^3 \rightarrow 12x^2$, $-9x^2 \rightarrow -18x$, $2 \rightarrow 0$
$f'''(x)$	$24x - 18$	Differentiate $12x^2 \rightarrow 24x$, $-18x \rightarrow -18$
$f^{(4)}(x)$	24	Constant
$f^{(5)}(x)$	0	Derivative of constant

Polynomials eventually vanish: A degree- n polynomial becomes zero after $n + 1$ derivatives.

Example 2: Physical Meaning — Position, Velocity, Acceleration, Jerk

Order	Math name	Physics name	What it measures
$f(t)$	position	position	Where you are
$f'(t)$	first derivative	velocity	How fast position changes
$f''(t)$	second derivative	acceleration	How fast velocity changes
$f'''(t)$	third derivative	jerk	How fast acceleration changes

A car moving with $s(t) = t^3 - 6t^2 + 9t$ (meters, seconds):

- $v(t) = s'(t) = 3t^2 - 12t + 9$ (velocity).
- $a(t) = s''(t) = 6t - 12$ (acceleration).
- At $t = 1$: $v = 0$ (stopped), $a = -6$ (decelerating — slowing down from positive direction).
- At $t = 3$: $v = 0$ (stopped again), $a = 6$ (accelerating — speeding up in positive direction).

Part B: Pattern Recognition — The n -th Derivative

The procedure: Compute the first 3–4 derivatives. Spot the repeating pattern. Write a formula for the n -th one.

Example 3: The Four Pattern Families

Family	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$	$f^{(n)}(x)$ formula
Exponential	e^{kx}	ke^{kx}	$k^2 e^{kx}$	$k^3 e^{kx}$	$k^n e^{kx}$
Sine	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin(x + \frac{n\pi}{2})$
Cosine	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos(x + \frac{n\pi}{2})$
Power (negative)	$1/x = x^{-1}$	$-x^{-2}$	$2x^{-3}$	$-6x^{-4}$	$(-1)^n n! x^{-(n+1)}$

Example 4: The Exponential Pattern — Multiply by k Each Time

$$f(x) = e^{3x}.$$

n	$f^{(n)}(x)$	Pattern
1	$3e^{3x}$	3^1
2	$9e^{3x}$	3^2
3	$27e^{3x}$	3^3
n	$3^n e^{3x}$	3^n

Check: $n = 5 \rightarrow 3^5 e^{3x} = 243e^{3x}$. Differentiate 5 times manually — it matches.

$f(x) = e^{-2x}$: $f^{(n)}(x) = (-2)^n e^{-2x}$. The sign alternates because the base is negative.

Example 5: The Trig Pattern — Cycle of 4

$f(x) = \sin x$:

n	0	1	2	3	4	5	6
$f^{(n)}(x)$	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$	$\cos x$	$-\sin x$

Procedure to find $f^{(n)}(x)$ for trig:

1. Divide n by 4. Keep the **remainder** r (0, 1, 2, or 3).
2. Read the answer from the cycle: $r = 0 \rightarrow \sin x$, $r = 1 \rightarrow \cos x$, $r = 2 \rightarrow -\sin x$, $r = 3 \rightarrow -\cos x$.

$f^{(100)}(\sin x)$: $100 \div 4 = 25$ remainder 0. $\rightarrow \sin x$. (Hundredth derivative returns to the start!)

$f^{(57)}(\sin x)$: $57 \div 4 = 14$ remainder 1. $\rightarrow \cos x$.

For $\cos x$: Same cycle offset: $r = 0 \rightarrow \cos x$, $r = 1 \rightarrow -\sin x$, $r = 2 \rightarrow -\cos x$, $r = 3 \rightarrow \sin x$.

Example 6: The Rational Pattern — Factorials Appear

$$f(x) = \frac{1}{x} = x^{-1}.$$

n	$f^{(n)}(x)$	Sign	Coefficient	Exponent
1	$-1 \cdot x^{-2}$	—	1	−2
2	$2 \cdot x^{-3}$	+	$2 = 2!$	−3
3	$-6 \cdot x^{-4}$	—	$6 = 3!$	−4
4	$24 \cdot x^{-5}$	+	$24 = 4!$	−5
n	$(-1)^n n! x^{-(n+1)}$	alternating	$n!$	$-(n + 1)$

General rational pattern for $\frac{1}{ax+b}$:

$$f^{(n)}(x) = (-1)^n \frac{a^n \cdot n!}{(ax+b)^{n+1}}.$$

$$f(x) = \frac{1}{2x+1}: f^{(3)}(x) = (-1)^3 \frac{2^3 \cdot 3!}{(2x+1)^4} = -\frac{48}{(2x+1)^4}.$$

Example 7: Polynomials — They Eventually Vanish

$$f(x) = x^5 - 2x^3 + x.$$

n	$f^{(n)}(x)$	Degree
1	$5x^4 - 6x^2 + 1$	4
2	$20x^3 - 12x$	3
3	$60x^2 - 12$	2
4	$120x$	1
5	120	0 (constant)

n	$f^{(n)}(x)$	Degree
6	0	— (vanished)

Rule: A degree- d polynomial becomes zero after $d + 1$ derivatives. The d -th derivative is constant: $f^{(d)}(x) = a_d \cdot d!$ (leading coefficient $\times$ factorial of degree).

Part C: The Leibniz Rule — n -th Derivative of a Product

Example 8: Leibniz Rule — The Product Rule Repeated

The pattern (looks like the binomial theorem):

$$(fg)'' = f''g + 2f'g' + fg''.$$

$$(fg)''' = f'''g + 3f''g' + 3f'g'' + fg'''.$$

General Leibniz rule:

$$(fg)^{(n)} = \sum_{k=0}^n \binom{n}{k} f^{(n-k)} g^{(k)}.$$

Coefficients are binomial coefficients: $\binom{n}{k} = \frac{n!}{k!(n-k)!}$.

When to use: One of the functions becomes zero after a few derivatives (e.g., polynomials — only a few non-zero terms).

$$f(x) = x^2 e^x. \quad n = 3:$$

- $f = x^2$: $f' = 2x$, $f'' = 2$, $f''' = 0$ (vanishes after 3 derivatives).
- $g = e^x$: $g' = g'' = g''' = e^x$ (never vanishes).

$$\begin{aligned} (x^2 e^x)''' &= \binom{3}{0} f''' g + \binom{3}{1} f'' g' + \binom{3}{2} f' g'' + \binom{3}{3} f g''' \\ &= 1 \cdot 0 \cdot e^x + 3 \cdot 2 \cdot e^x + 3 \cdot 2x \cdot e^x + 1 \cdot x^2 \cdot e^x \end{aligned}$$

$$= e^x(6 + 6x + x^2).$$

Part D: Implicit Second Derivatives

Example 9: Finding $\frac{d^2y}{dx^2}$ from an Implicit Equation

Procedure:

Step	Action
1	Differentiate the equation implicitly to find $\frac{dy}{dx}$ (as in 14B)
2	Differentiate $\frac{dy}{dx}$ with respect to x — this gives $\frac{d^2y}{dx^2}$
3	Substitute the expression for $\frac{dy}{dx}$ from step 1 into the result
4	The answer will involve x and y — that's expected for implicit curves

$x^2 + y^2 = 25$. Find $\frac{d^2y}{dx^2}$.

Step 1: $2x + 2y \frac{dy}{dx} = 0 \rightarrow \frac{dy}{dx} = -\frac{x}{y}$.

Step 2: Differentiate $\frac{dy}{dx} = -\frac{x}{y}$ with respect to x . Use quotient rule:

$$\frac{d^2y}{dx^2} = \frac{d}{dx} \left(-\frac{x}{y} \right) = -\frac{1 \cdot y - x \cdot \frac{dy}{dx}}{y^2} = -\frac{y - x \frac{dy}{dx}}{y^2}.$$

Step 3: Substitute $\frac{dy}{dx} = -\frac{x}{y}$:

$$\frac{d^2y}{dx^2} = -\frac{y - x(-\frac{x}{y})}{y^2} = -\frac{y + \frac{x^2}{y}}{y^2} = -\frac{\frac{y^2 + x^2}{y}}{y^2} = -\frac{y^2 + x^2}{y^3}.$$

Step 4: Since $x^2 + y^2 = 25$, the numerator simplifies: $-\frac{25}{y^3}$.

Geometric meaning: On the circle $x^2 + y^2 = 25$, the second derivative is always negative when $y > 0$ (concave down — the top half of the circle curves downward) and positive when $y < 0$ (concave up).

Part E: Concavity and Inflection Points

Example 10: Reading Concavity from f''

Sign of $f''(x)$	Shape	Meaning
$f''(x) > 0$	Concave up $\smile$	Slope is increasing — graph bends upward
$f''(x) < 0$	Concave down $\frown$	Slope is decreasing — graph bends downward
$f''(x) = 0$	Possible inflection	Curvature may switch — test sign change

Procedure to find inflection points:

1. Compute $f''(x)$.
2. Set $f''(x) = 0$ and solve for x .
3. Check if f'' **changes sign** at those x -values. If yes $\rightarrow$ inflection point. If no $\rightarrow$ not an inflection.

$$f(x) = x^3 - 3x: f'(x) = 3x^2 - 3, f''(x) = 6x.$$

$$f''(x) = 0 \text{ at } x = 0.$$

- $x < 0$: $f''(x) < 0$ (concave down).
- $x > 0$: $f''(x) > 0$ (concave up).

Sign changes $\rightarrow (0, 0)$ is an **inflection point**.

$f(x) = x^4$: $f''(x) = 12x^2$. $f''(0) = 0$, but $f''(x) \geq 0$ everywhere (never changes sign). No inflection — it's always concave up.

Example 11: Mixed Technique Problems

Procedure for mixed problems: Identify the outer structure first (product? quotient? chain?),

apply the matching rule from 14B. Then differentiate again if needed.

$$f(x) = e^x \sin x + \ln(\cos x):$$

1. Split at the $+$: two terms.
2. Term 1 — product rule: $(e^x)' \sin x + e^x (\sin x)' = e^x \sin x + e^x \cos x = e^x (\sin x + \cos x)$.
3. Term 2 — chain rule: outer $\ln(\square) \rightarrow 1/\square$, inner $\cos x \rightarrow -\sin x$. $= \frac{-\sin x}{\cos x} = -\tan x$.
4. $f'(x) = e^x (\sin x + \cos x) - \tan x$.

$$f(x) = \arctan(\ln x):$$

1. Chain rule: outer $\arctan(\square) \rightarrow \frac{1}{1+\square^2}$, inner $\ln x \rightarrow 1/x$.
2. $f'(x) = \frac{1}{1+(\ln x)^2} \cdot \frac{1}{x}$.

$$f(x) = x^3 e^{2x} \tan x \text{ (triple product):}$$

$$(fgh)' = f'gh + fg'h + fgh'.$$

1. $f'gh = 3x^2 e^{2x} \tan x$.
2. $fg'h = x^3 \cdot 2e^{2x} \cdot \tan x$.
3. $fgh' = x^3 e^{2x} \cdot \sec^2 x$.
4. Factor $x^2 e^{2x}$: $x^2 e^{2x} [3 \tan x + 2x \tan x + x \sec^2 x]$.

Up to here: Higher derivatives = differentiate repeatedly. Exponential: multiply by k each time. Trig: cycle of 4. Rational: $(-1)^n n! x^{-(n+1)}$. Polynomial: vanishes after $d + 1$ derivatives. Leibniz rule: $(fg)^{(n)} = \sum \binom{n}{k} f^{(n-k)} g^{(k)}$. Implicit 2nd derivative: differentiate dy/dx again, substitute. Concavity: $f'' > 0 \rightarrow \text{up}$, $f'' < 0 \rightarrow \text{down}$. Inflection: $f'' = 0$ + sign change.

Common Mistakes

Mistake 1: Stopping the trig cycle at the wrong remainder

Wrong: " $f^{(7)}(\sin x)$: $7 \div 4$ remainder 3 $\rightarrow \sin x$." **Right:** Remainder 0 = $\sin x$, 1 = $\cos x$, 2 = $-\sin x$, 3 = $-\cos x$. Count carefully from 0.

Mistake 2: Forgetting the factorial in rational n -th derivatives

Wrong: $f^{(n)}(1/x) = (-1)^n x^{-(n+1)}$. **Right:** $(-1)^n n! x^{-(n+1)}$. The coefficient grows factorially — test with $n = 2$: $f''(1/x) = 2x^{-3}$, not x^{-3} .

Mistake 3: Claiming $f''(x) = 0$ always means inflection

Wrong: " $f(x) = x^4$, $f''(0) = 0$, so $(0, 0)$ is an inflection point." **Right:** f'' must CHANGE SIGN. $f''(x) = 12x^2 \geq 0$ always — no sign change, no inflection.

Mistake 4: Forgetting to substitute dy/dx in implicit second derivatives

Wrong: Leaving $\frac{dy}{dx}$ in the final answer for $\frac{d^2y}{dx^2}$. **Right:** Substitute the first derivative expression back in. The final answer should only contain x and y .

What We Just Did

(1) Higher derivatives: differentiate repeatedly. $f'' = (f')'$, $f''' = (f'')'$, etc.
Physical: f' =velocity, f'' =acceleration, f''' =jerk.

(2) n th-derivative patterns:

Exponential: $f^{(n)}(e^{kx}) = k^n e^{kx}$.

Trig: cycle of 4 — use remainder of $n \div 4$.

Rational: $f^{(n)}(1/x) = (-1)^n n! x^{-(n+1)}$.

Polynomial of degree d : vanishes after $d+1$ derivatives.

(3) Leibniz rule: $(fg)^{(n)} = \sum \text{binom}(n, k) f^{(n-k)} g^{(k)}$. Best when one function vanishes.

(4) Implicit 2nd derivative: find dy/dx , differentiate again, substitute dy/dx .

(5) Concavity: $f'' > 0 \rightarrow \text{up}$, $f'' < 0 \rightarrow \text{down}$. Inflection: $f'' = 0$ AND sign change.

Practice 1

Find $f''(x)$ for $f(x) = x^5 - 2x^3 + x$. Show each step clearly.

→ Reference: **Example 1**

Solutions: [Solutions](#)

Practice 2

Find $f^{(4)}(x)$ for $f(x) = e^{2x}$. Use the exponential pattern: $k^n e^{kx}$.

→ Reference: **Example 4**

Solutions: [Solutions](#)

Practice 3

Find $f^{(57)}(\cos x)$. Use the trig cycle: divide by 4, take the remainder.

→ Reference: **Example 5**

Solutions: [Solutions](#)

Practice 4

Find $f^{(n)}(x)$ for $f(x) = \frac{1}{2x+1}$. Use the rational pattern formula.

→ Reference: **Example 6**

Solutions: [Solutions](#)

Practice 5

Find $\frac{d^2y}{dx^2}$ for $x^3 + y^3 = 6xy$. (Implicit second derivative — find dy/dx first, then differentiate again.)

→ Reference: **Example 9**

Solutions: [Solutions](#)

Practice 6: Real Battle (Constructive)

$f(x) = x^3 - 3x^2 - 9x + 5$. (a) Find all x where $f'(x) = 0$ (critical points). (b) Find all x where $f''(x) = 0$ (possible inflection). (c) Determine intervals where f is concave up and concave down. (d) Sketch the concavity behavior — where does the graph bend upward vs. downward?

→ Reference: **Example 10**

Solutions: [Solutions](#)

Basic Algebra Drill — Higher Derivatives (10 Problems)

Differentiate repeatedly. Spot the pattern for n -th derivatives.

D1. Find $f''(x)$ for $f(x) = 3x^4 - 5x^2 + 2x - 7$. Differentiate twice.

D2. Find $f'''(x)$ for $f(x) = x^5$. Three derivatives of a monomial.

D3. Find $f''(x)$ for $f(x) = e^{-x}$. Chain rule in first derivative, then again.

D4. Find $f''(x)$ for $f(x) = \ln x$. First = $1/x$, second = differentiate $1/x$.

- D5.** Find $f''(x)$ for $f(x) = \sin 2x$. Chain rule: $\cos 2x \cdot 2$, then differentiate again.
- D6.** Find $f^{(n)}(x)$ for $f(x) = e^{5x}$. Pattern: $k^n e^{kx}$.
- D7.** Find $f''(x)$ for $f(x) = x \ln x$. Product rule, then differentiate again.
- D8.** Find $f''(0)$ for $f(x) = \tan x$. $f'(x) = \sec^2 x$, $f''(x) = 2 \sec^2 x \tan x$.
- D9.** $f(x) = |x^3|$. Where is $f'(x)$ undefined? Where is $f''(x)$ undefined?
- D10.** Find the inflection point(s) of $f(x) = x^3 - 6x^2 + 9x$. Compute f'' , set to zero, check sign change.

Solutions: [Solutions](#)

Advanced Algebra Drill — Higher Derivatives (10 Problems)

Patterns, proofs, implicit, and Leibniz.

- A1.** Prove Leibniz rule for $n = 2$: $(fg)'' = f''g + 2f'g' + fg''$. Start from $(fg)' = f'g + fg'$, differentiate again.
- A2.** Find $f^{(100)}(x)$ for $f(x) = xe^x$. Use Leibniz rule — x vanishes after 2 derivatives, so only 2 terms survive.
- A3.** $f(x) = \frac{1}{1-x}$. Find a formula for $f^{(n)}(x)$. Compute f' , f'' , f''' , spot the pattern.
- A4.** Find $f''(\pi/4)$ for $f(x) = \sin^2 x$. Use $\sin^2 x = \frac{1 - \cos 2x}{2}$ first, OR use chain rule twice.
- A5.** $f(x) = \ln(\sin x + \cos x)$. Find $f'(x)$, then $f''(0)$. Quotient rule in the second derivative.
- A6.** Prove $y = e^x \sin x$ satisfies $y'' - 2y' + 2y = 0$. Compute y' , y'' , plug in, simplify.
- A7.** Find all x where $f''(x) = 0$ for $f(x) = x^4 - 6x^2 + 8x$. Compute f'' , solve quadratic, verify sign change.
- A8.** Find $\frac{d^2y}{dx^2}$ for $x^2 + xy + y^2 = 7$ using implicit differentiation twice. (Product rule on xy !)

A9. $f(x) = x^{x^x}$. Find $f'(x)$ using log-diff twice. Then find $f''(1)$. (Challenging — take $\ln$ twice.)

A10. $f(x) = \frac{ax+b}{cx+d}$. Prove $f'''(x) = 0$ for all $x \neq -d/c$. (Rewrite as $A + \frac{B}{cx+d}$ using division — then only one term to differentiate.)

Solutions: [Solutions](#)

Today's Procedure

When you see...	Do this...
Find f'' , f''' , etc.	Differentiate repeatedly. No new rules — just persistence
Find $f^{(n)}(e^{kx})$	Formula: $k^n e^{kx}$. Multiply by k each time
Find $f^{(n)}(\sin x)$ or $f^{(n)}(\cos x)$	Divide n by 4. Remainder tells you where in the cycle
Find $f^{(n)}(1/(ax+b))$	Formula: $(-1)^n a^n n! / (ax+b)^{n+1}$
Find n -th derivative of a product	Leibniz rule: $\sum \binom{n}{k} f^{(n-k)} g^{(k)}$. Best when one factor vanishes
Find $\frac{d^2 y}{dx^2}$ from implicit equation	Find dy/dx first, differentiate it again, substitute dy/dx back
Determine concavity	Compute f'' . Positive $\rightarrow$ up. Negative $\rightarrow$ down
Find inflection points	Set $f'' = 0$, solve. Check that f'' CHANGES SIGN
Mixed techniques (product+chain+trig)	Identify outer structure $\rightarrow$ apply matching 14B rule $\rightarrow$ differentiate again if needed

Terminology

What we call it	Math term	Notation
derivative of the derivative	second derivative	$f''(x), \frac{d^2y}{dx^2}$
rate of change of rate of change	third derivative / jerk	$f'''(x)$
n -times differentiated	n -th derivative	$f^{(n)}(x), \frac{d^ny}{dx^n}$
bends upward	concave up	$f''(x) > 0$
bends downward	concave down	$f''(x) < 0$
curvature switches	inflection point	$f''(x) = 0 + \text{sign change}$
n -th derivative of product	Leibniz rule	$(fg)^{(n)} = \sum \binom{n}{k} f^{(n-k)} g^{(k)}$
trig cycle length 4	periodic derivative pattern	$\sin \rightarrow \cos \rightarrow -\sin \rightarrow -\cos$
polynomial eventually zero	finite differences	degree- d vanishes at $d + 1$